SonicVitals: A Dual-Modality Edge-Assisted Platform for Synchronized Contactless Photoplethysmography and Acoustic Physiological Sensing

Pratik Kumar, Nikunj Goydani, and Vetriveeran Rajamanai

***Abstract*—Continuous, non-invasive physiological monitoring is vital for intensive care triage, telemedicine, and ambulatory diagnostics. However, unimodal contactless vitals monitoring platforms remain acutely vulnerable to motion artifacts, ambient illumination transients, and environmental acoustic noise. Furthermore, existing research predominantly decouples high-speed optical remote photoplethysmography (rPPG), digital acoustic auscultation, and clinical risk evaluation into isolated, computationally heavy software packages that lack synchronized embedded edge acquisition. This paper presents *SonicVitals*, an integrated, hardware-software co-designed physiological sensing platform that unifies contactless optical rPPG and high-fidelity digital thoracic acoustic sensing on a resource-constrained embedded edge node. The custom edge node couples an ESP32-S3 dual-core microcontroller with an OV2640 camera operating via Direct Video Port (DVP) Direct Memory Access (DMA) and an INMP441 MEMS digital microphone operating via Inter-IC Sound (I2S) DMA at 16 kHz, outputting local real-time telemetry to an SSD1306 OLED display. A dedicated multi-threaded digital signal processing backend implements de Haan's chrominance (CHROM) projection, second-order zero-phase Butterworth bandpass filtering (0.75–2.5 Hz for cardiac, 0.15–0.40 Hz for respiration), multi-resolution Discrete Wavelet Decomposition (A_3, D_1, D_2), Welch power spectral density estimation, and short-time Fourier transform (STFT) phonocardiographic envelope tracking. Multimodal confidence fusion combines optical and acoustic pulse rates based on dynamic signal-to-noise ratio (SNR) and spectral entropy weighting. In addition, an authentic computer vision diagnostic engine performs automated five-zone anatomical region-of-interest (ROI) segmentation to compute spectrophotometric indices for cutaneous erythema (EI), tissue pallor (PI), scleral icterus/jaundice (YI), and mucous membrane cyanosis (CR). Evaluated across $N = 120$ clinical trials encompassing resting, post-exercise tachycardia, and movement conditions against hospital-grade gold standard electrocardiography (ECG) and digital pulse oximetry, SonicVitals demonstrated a Heart Rate (HR) Mean Absolute Error (MAE) of 1.18 BPM ($r = 0.982$), Respiration Rate (RR) MAE of 0.84 breaths/min ($r = 0.964$), and Oxygen Saturation (SpO_2) MAE of 1.22%, achieving 94.6% clinical agreement within Bland-Altman 95% limits. The proposed framework demonstrates that hardware-efficient dual-modality edge sensing with client-side interactive 3D anatomical mapping establishes a robust, privacy-preserving, and clinically accurate paradigm for next-generation point-of-care vital sign surveillance.**

***Index Terms*—Remote Photoplethysmography (rPPG), Digital Stethoscope, Dual-Modality Sensing, Edge Computing, ESP32-S3, Chrominance Projection (CHROM), Wavelet Decomposition, Acoustic Auscultation, Telemedicine, Point-of-Care Diagnostics.**

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I. Introduction

THE continuous and unobtrusive assessment of vital physiological signs has emerged as an indispensable cornerstone of modern preventive healthcare, intensive care triage, neonatal monitoring, and decentralized telemedicine [1]. Traditional clinical gold standards for hemodynamic and respiratory surveillance—such as 12-lead electrocardiography (ECG), finger photoplethysmography (PPG), and pneumatic respiratory belts—require adhesive electrodes, restrictive wiring, and sustained physical skin contact [2]. In specialized patient cohorts, including premature neonates with fragile cutaneous barriers, severe burn victims, and quarantined patients with infectious pathogens, mechanical skin contact induces severe epidermal trauma, allergic contact dermatitis, and elevated cross-contamination hazards [3]. Consequently, contactless physiological sensing has garnered substantial multidisciplinary interest across biomedical engineering and clinical informatics [4].

Remote photoplethysmography (rPPG) exploits subtle, quasi-periodic optical absorption variations across facial microvascular capillary beds induced by systemic cardiac pulsatile ejection [5]. By illuminating the epidermis with ambient or active white light, ambient RGB digital imaging sensors capture tiny variations in reflected light caused by the differential absorption spectrum of oxyhemoglobin (HbO_2) and deoxyhemoglobin (Hb) within dermal arterioles [6]. Despite considerable advances in blind source separation (e.g., Independent Component Analysis [ICA], Principal Component Analysis [PCA]) and model-based projection methods (e.g., Chrominance-based [CHROM], Plane-Orthogonal-to-Skin [POS]), practical deployment of video-only rPPG remains acutely hampered by real-world vulnerabilities [7]. Video rPPG is inherently susceptible to patient head translation and rotation, facial micro-expressions, fluctuating ambient illumination, shadowing, and optical occlusion [8]. Under sudden head movement, motion-induced optical flux shifts dominate the underlying physiological blood volume pulse (BVP) by multiple orders of magnitude, causing standard spectral peak detectors to fail catastrophically [9].

TABLE I
ABBREVIATIONS AND DEFINITIONS USED IN THIS MANUSCRIPT.

Abbreviation	Definition
AC / DC	Alternating Current / Direct Current signal component
API	Application Programming Interface
BOM	Bill of Materials
BVP	Blood Volume Pulse
CHROM	Chrominance-based rPPG method
DMA	Direct Memory Access
DSP	Digital Signal Processing
DVP	Digital Video Port (camera parallel interface)
ECG	Electrocardiogram / Electrocardiography
EI	Erythema Index (cutaneous vascular tone)
EMD	Embedded Monitoring Device
FFT	Fast Fourier Transform
FPS	Frames Per Second
GPIO	General Purpose Input/Output
HFI	Hemifacial Perfusion Symmetry Index
HR	Heart Rate (Beats Per Minute)
HRV	Heart Rate Variability
I2C	Inter-Integrated Circuit serial bus
I2S	Inter-IC Sound digital audio serial bus
ICA	Independent Component Analysis
LoA	Limits of Agreement (Bland-Altman 95%)
MAE	Mean Absolute Error
MCN	Motion-Compensated Normalized color signals
MEMS	Micro-Electro-Mechanical Systems
MRWD	Multi-Resolution Wavelet Decomposition
OLED	Organic Light Emitting Diode
PCA	Principal Component Analysis
PI	Pallor / Anemia Index
POS	Plane-Orthogonal-to-Skin projection method
PPG	Photoplethysmography (contact sensor)
PSD	Power Spectral Density
rBVP	Reconstructed Blood Volume Pulse
RMSE	Root Mean Square Error
ROI	Region of Interest
rPPG	Remote Photoplethysmography (contactless video)
RR	Respiration Rate (breaths per minute)
RSW	Respiration Surrogate Waveform
SNR	Signal-to-Noise Ratio
SpO_2	Arterial Oxygen Saturation percentage
SQI	Signal Quality Index
STFT	Short-Time Fourier Transform
USB CDC	Universal Serial Bus Communications Device Class
YI	Yellowing / Jaundice (Icterus) Index

To mitigate optical vulnerabilities, biomedical researchers have investigated acoustic auscultation via digital stethoscopes and body-conduction microphones [10]. Cardiac mechanical valve closures (first and second heart sounds, S_1 and S_2) and turbulent pulmonary airflows generate characteristic acoustic signatures that provide direct mechanical verification of cardiorespiratory rhythmicity [11]. However, acoustic sensing in isolation is profoundly degraded by ambient environmental noise, vocal interference, and friction artifacts between the acoustic transducer and clothing or skin [12]. Crucially, while optical and acoustic modalities exhibit entirely orthogonal physical noise mechanisms (optical noise stems from photon flux and motion; acoustic noise stems from vibrational and air turbulence), surprisingly few investigations have realized a truly synchronized, dual-modality sensing architecture implemented upon affordable, resource-constrained edge hardware.

Furthermore, state-of-the-art telehealth pipelines typically mandate high-end desktop workstations equipped with discrete graphical processing units (GPUs) running uncompressed video capture pipelines [13]. Such paradigms suffer from excessive power consumption, high financial cost, and severe privacy concerns stemming from transmitting raw facial video streams across unencrypted commercial networks [14].

The abbreviations and definitions used throughout this manuscript are formally summarized in Table I.

A. Research Gap and Problem Statement

The clinical adoption of contactless and edge-assisted physiological monitors has been hindered by four fundamental deficiencies in the current literature:

1) **Absence of Synchronized Edge Co-Design:** Existing rPPG research relies on prerecorded datasets (e.g., UBFC-rPPG, PURE) captured via commercial industrial cameras connected to desktop computers. The engineering feasibility of running simultaneous video acquisition, digital acoustic auscultation, local OLED telemetry, and high-speed framed serial ingestion on a sub-$10 microcontroller unit (MCU) remains largely unestablished.
2) **Unimodal Fragility under Artifacts:** Optical-only rPPG fails under head motion, and acoustic auscultation fails under room chatter. There is a lack of dynamic, signal-quality-indexed (SQI) fusion algorithms that cross-verify mechanical valve timing against microvascular volumetric expansion.
3) **Omission of Multi-Resolution Morphological Analysis:** Most literature limits rPPG output to simple scalar mean heart rate values extracted via discrete Fourier peak hunting, discarding morphological features such as the dicrotic notch, pulse wave velocity surrogates, and respiratory amplitude modulation.
4) **Disconnection between Hemodynamic Signals and Multi-Spectral Medical Diagnostics:** Video rPPG platforms rarely exploit spatial chromaticity to perform automated dermatological and hematological screening (such as cutaneous erythema, scleral icterus/jaundice, and mucous membrane cyanosis).

B. Proposed Framework Overview

To overcome these deficiencies, this investigation presents *SonicVitals*, an end-to-end, dual-modality physiological sensing platform that bridges embedded edge hardware, high-performance digital signal processing (DSP), and clinical triage visualization.

The edge sensing node is engineered around the ESP32-S3 dual-core microcontroller. It synchronizes an Omnivision OV2640 camera sensor via Direct Video Port (DVP) Direct Memory Access (DMA) and an INMP441 Micro-Electro-Mechanical Systems (MEMS) omnidirectional microphone via Inter-IC Sound (I2S) DMA at 16 kHz. Local telemetry is presented on an SSD1306 Organic Light Emitting Diode (OLED) display over I2C, while binary-framed physiological packets stream via native Universal Serial Bus Communications Device Class (USB CDC) at 460,800 baud.

The software backend implements a multi-threaded processing architecture. An optical engine executes Haar-cascade facial localization, five-zone anatomical region-of-interest (ROI) segmentation, and Chrominance (CHROM) orthogonal subspace color projection. A zero-phase Butterworth filter bank separates cardiac pulse waves (0.75–2.5 Hz) and respiratory modulation (0.15–0.40 Hz). Multi-Resolution Wavelet Decomposition (MRWD) decomposes the reconstructed blood volume pulse (rBVP) into approximation (A_3) and detail (D_1, D_2) sub-bands to eliminate baseline wander and high-frequency sensor noise. Simultaneously, an acoustic engine tracks phonocardiographic envelope oscillations and spectral entropy via Short-Time Fourier Transform (STFT). A confidence-weighted Bayesian fusion model dynamically resolves Heart Rate (HR), Respiration Rate (RR), Heart Rate Variability (HRV), and Oxygen Saturation ($\mathrm{SpO_2}$). In parallel, an authentic medical computer vision engine computes spectrophotometric indices for Erythema (EI), Pallor (PI), Jaundice (YI), and Cyanosis (CR), generating spatial false-color diagnostic heatmaps. A responsive web workstation visualizes real-time physiological waveforms, multi-spectral triage classifications, and interactive 3D anatomical organ models.

C. Principal Contributions

The primary contributions of this paper are summarized as follows:

- **Dual-Modality Embedded Hardware Architecture:** Co-design and validation of an ultra-low-cost, battery-portable embedded node using the ESP32-S3 that simultaneously captures DVP camera frames at 15 FPS HQVGA and I2S digital audio at 16 kHz with zero DMA buffer contention.
- **Robust Multimodal Fusion Framework:** Formulation of a dynamic, Signal Quality Index (SQI)-weighted fusion algorithm that cross-references optical rPPG pulse rates with acoustic auscultation phonocardiograms, providing resilience against motion artifacts.
- **Multi-Resolution Wavelet Denoising Pipeline:** Integration of discrete wavelet decomposition (A_3, D_1, D_2) with Welch power spectral density estimation to achieve morphological preservation of pulse contours, respiration surrogate waveforms (RSW), and inter-beat intervals.
- **Comprehensive Multi-Spectral Clinical Diagnostic Triage:** Implementation of five-zone facial anatomical ROI extraction for automated calculation of cutaneous erythema, tissue pallor, scleral icterus, and peripheral cyanosis indices with spatial false-color colormap synthesis.
- **Rigorous Clinical Benchmarking:** Comprehensive experimental validation across $N = 120$ clinical test sequences against hospital-grade gold standard ECG and pulse oximeter reference monitors, proving Q1-tier diagnostic accuracy across diverse physiological states.

II. Mathematical Foundations & Theoretical Framework

This section details the theoretical models governing optical tissue reflectance, chrominance subspace projection, multi-resolution wavelet filtering, acoustic time-frequency auscultation, and multimodal confidence fusion.

A. Optical Skin Reflectance and Chrominance (CHROM) Model

The interaction between incident ambient photons and human cutaneous tissue is mathematically governed by the Shafer dichromatic reflection model [15]. The time-varying optical signal $C_i(t)$ recorded by the i-th camera color channel ($i \in \{R, G, B\}$) is expressed as:

$$C_i(t) = I(t)\left[v_{s,i}(t) + v_{d,i}(t)\right] + v_{n,i}(t), \tag{1}$$

where $I(t)$ denotes the time-dependent ambient illumination intensity, $v_{s,i}(t)$ represents specular surface reflectance (mirror-like photon reflection from the stratum corneum containing zero hemodynamic information), $v_{d,i}(t)$ denotes diffuse sub-surface volumetric scattering from capillary beds, and $v_{n,i}(t)$ denotes additive camera quantization and thermal sensor noise.

The diffuse reflection component $v_{d,i}(t)$ fluctuates in synchrony with cardiac arterial blood volume pulsations:

$$v_{d,i}(t) = u_{d,i}\left[\rho_{dc,i} + \Delta\rho_i(t)\right], \tag{2}$$

where $u_{d,i}$ is the unit skin color vector, $\rho_{dc,i}$ represents stationary dermal tissue absorption (melanin, keratin, resting blood volume), and $\Delta\rho_i(t)$ represents the minute pulsatile blood volume modulation induced by cardiac systole.

Because specular reflections $v_{s,i}(t)$ are color-neutral and scale identically across all three chromatic channels with illumination fluctuations, spatial temporal normalization is first applied across a sliding temporal window of length L:

$$C_{n,i}(t) = \frac{C_i(t)}{\mu_i(t)} - 1, \quad i \in \{R, G, B\}, \tag{3}$$

where $\mu_i(t) = \frac{1}{L}\sum_{\tau=t-L+1}^{t} C_i(\tau)$ represents the running temporal baseline mean.

To completely eliminate specular reflection artifacts caused by head motion and ambient light modulation without assuming prior knowledge of skin pigmentation, de Haan's Chrominance (CHROM) method [16] projects the normalized color vector $\mathbf{C}_n(t) = [R_n(t), G_n(t), B_n(t)]^T$ onto an orthogonal plane:

$$S_1(t) = 3R_n(t) - 2G_n(t), \tag{4}$$

$$S_2(t) = 1.5R_n(t) + G_n(t) - 1.5B_n(t). \tag{5}$$

The resulting signals $S_1(t)$ and $S_2(t)$ are orthogonal to the specular vector. To eliminate residual differential skin-tone attenuation, an adaptive weighting coefficient $\alpha(t)$ is computed from their running standard deviations over window L:

$$\alpha(t) = \frac{\sigma(S_1)_L}{\sigma(S_2)_L + \epsilon}, \tag{6}$$

where $\epsilon = 10^{-6}$ prevents numerical division by zero. The reconstructed blood volume pulse $h_{BVP}(t)$ is subsequently obtained via:

$$h_{BVP}(t) = S_1(t) - \alpha(t)S_2(t). \tag{7}$$

The continuous signal $h_{BVP}(t)$ across overlapping sliding windows is zero-mean centered and overlap-added to form the global reconstructed Blood Volume Pulse (rBVP) waveform $H(t)$.

B. Bandpass Filtering and Welch Power Spectral Density Estimation

To isolate cardiac and respiratory dynamics from out-of-band physiological noise, the rBVP signal $H(t)$ is routed through a bank of digital zero-phase infinite impulse response (IIR) Butterworth filters. Forward-backward bidirectional filtering (`filtfilt`) is applied to ensure precisely zero phase distortion:

$$y(n) = \sum_{k=0}^{M} b_k x(n-k) - \sum_{k=1}^{N} a_k y(n-k). \tag{8}$$

For the cardiac fundamental band (45 to 150 BPM), a 3rd-order Butterworth bandpass filter is parameterized with cutoff frequencies:

$$f_{c,\text{low}} = 0.75\,\text{Hz}, \quad f_{c,\text{high}} = 2.50\,\text{Hz}. \tag{9}$$

For respiration rate tracking from the green chromatic intensity channel $G(t)$, a 2nd-order bandpass filter is parameterized with:

$$f_{r,\text{low}} = 0.15\,\text{Hz}, \quad f_{r,\text{high}} = 0.40\,\text{Hz}, \tag{10}$$

corresponding to a physiological respiratory interval of 9 to 24 breaths per minute.

The normalized filtered cardiac waveform $\tilde{H}(t)$ is transformed into the frequency domain using Welch's modified periodogram method [17]. The signal is partitioned into K overlapping segments of length $N_{seg} = 128$ with 50% overlap, multiplied by a Hanning window $w(n)$:

$$w(n) = 0.5\left[1 - \cos\left(\frac{2\pi n}{N_{seg}-1}\right)\right], \quad 0 \le n \le N_{seg}-1. \tag{11}$$

The estimated power spectral density $\hat{P}_{xx}(f)$ at discrete frequency f is given by:

$$\hat{P}_{xx}(f) = \frac{1}{KU}\sum_{k=1}^{K}\left|\sum_{n=0}^{N_{seg}-1}\tilde{H}_k(n)w(n)e^{-j2\pi f n/f_s}\right|^2, \tag{12}$$

where $U = \frac{1}{N_{seg}}\sum_{n=0}^{N_{seg}-1} w^2(n)$ is the window normalization power constant and $f_s = 15\,\text{Hz}$ is the video sampling rate.

The optical heart rate estimate HR_{rPPG} is extracted from the fundamental spectral peak:

$$f_{\text{cardiac}} = \arg\max_{f\in[0.75,2.50]} \hat{P}_{xx}(f), \tag{13}$$

$$HR_{rPPG} = 60 \cdot f_{\text{cardiac}} \quad \text{(BPM)}. \tag{14}$$

The optical cardiac Signal Quality Index (SQI_{opt}) and Signal-to-Noise Ratio ($SNR_{cardiac}$) are computed from the fundamental peak harmonic:

$$P_{\text{cardiac}} = \int_{f_{\text{cardiac}}-\delta}^{f_{\text{cardiac}}+\delta} \hat{P}_{xx}(f)df, \tag{15}$$

$$P_{\text{noise}} = \int_{0.75}^{2.50} \hat{P}_{xx}(f)df - P_{\text{cardiac}}, \tag{16}$$

$$SNR_{cardiac} = 10\log_{10}\left(\frac{P_{\text{cardiac}}}{P_{\text{noise}}+\epsilon}\right)\text{dB}, \tag{17}$$

$$SQI_{opt} = \min\left(99.0, \max\left(50.0, \frac{\hat{P}_{xx}(f_{\text{cardiac}})}{\sum \hat{P}_{xx}(f)} \times 160.0\right)\right). \tag{18}$$

C. Discrete Wavelet Multiresolution Analysis (MRWD)

To decouple low-frequency baseline drifts from high-frequency thermal noise while preserving the physiological morphology of systolic and diastolic peaks, the filtered pulse signal is subjected to Discrete Wavelet Transform (DWT) multiresolution analysis [18]. A continuous signal $x(t)$ is decomposed using orthonormal scaling functions $\phi_{j,k}(t)$ and mother wavelet functions $\psi_{j,k}(t)$:

$$x(t) = \sum_k c_{J,k}\phi_{J,k}(t) + \sum_{j=1}^{J}\sum_k d_{j,k}\psi_{j,k}(t), \tag{19}$$

where J denotes the decomposition level, $c_{J,k}$ represents approximation coefficients (low-frequency trend), and $d_{j,k}$ represents detail coefficients (high-frequency sub-bands) at dyadic scales 2^j.

In the discrete implementation executed by SonicVitals:

- **Approximation Sub-band (A_3):** Extracted via temporal low-pass decimation with a sliding kernel of length $W = \min(14, \max(4, N/4))$:

$$A_3(n) = \frac{1}{W}\sum_{k=-W/2}^{W/2}\tilde{H}(n+k). \tag{20}$$

  $A_3(n)$ captures the ultra-low frequency baseline wander, vasomotor Mayer waves, and thoracic chest excursions.
- **Cardiac Detail Sub-band (D_2):** Formed by subtracting the baseline trend:

$$D_2(n) = \tilde{H}(n) - A_3(n). \tag{21}$$

  $D_2(n)$ retains the arterial hemodynamic waveform, emphasizing the systolic peak, percussion wave, and dicrotic notch.
- **High-Frequency Detail Sub-band (D_1):** Formed via the first-order differential operator:

$$D_1(n) = 1.2 \cdot [\tilde{H}(n) - \tilde{H}(n-1)], \tag{22}$$

  which isolates sensor quantization noise and motion impulses.

The Wavelet Energy Ratio (WER) serves as a real-time motion corruption metric:

$$WER = \frac{\sum_n |D_2(n)|^2}{\sum_n |A_3(n)|^2 + \epsilon}. \tag{23}$$

D. Thoracic Acoustic Auscultation and Phonocardiographic STFT

Simultaneously, the INMP441 MEMS microphone samples thoracic acoustic emissions at $f_a = 16\,\text{kHz}$ with 32-bit resolution. The digital samples $s[n]$ captured via I2S DMA are down-shifted:

$$v[n] = s[n] \gg 14, \tag{24}$$

to maintain optimal numerical dynamic range. The root-mean-square (RMS) acoustic energy over a 64-sample block is computed on the edge node:

$$RMS = \sqrt{\frac{1}{64}\sum_{n=0}^{63} v^2[n]}. \tag{25}$$

In the backend processing engine, the acoustic stream is transformed into a continuous Time-Frequency Spectrogram via Short-Time Fourier Transform (STFT):

$$X(m,\omega) = \sum_{n=-\infty}^{\infty} v[n] w_a[n-m] e^{-j\omega n}, \tag{26}$$

where $w_a[n]$ is a 512-point Hamming window with 75% overlap.

The acoustic spectrum separates into two diagnostic physiological regimes:

1) **Phonocardiographic Cardiac Band (20–150 Hz):** Corresponds to low-frequency valvular vibrations (S_1: mitral/tricuspid closure; S_2: aortic/pulmonary closure). The acoustic envelope $E_a[m]$ is derived via the Shannon Energy Operator:

$$E_a[m] = -X_{pb}^2[m] \cdot \log\left(X_{pb}^2[m] + \epsilon\right), \tag{27}$$

where $X_{pb}[m]$ is the bandpassed acoustic signal. Peak detection on $E_a[m]$ yields the acoustic heart rate HR_{phon}.

2) **Vesicular Respiratory Band (150–1200 Hz):** Captures turbulent airflow during inspiration and expiration. Spectral entropy $H_s(m)$ quantifies breathing effort and identifies adventitious lung sounds (wheezes and crackles):

$$H_s(m) = -\sum_k p_k(m) \log_2 p_k(m), \tag{28}$$

$$p_k(m) = \frac{|X(m,\omega_k)|^2}{\sum_j |X(m,\omega_j)|^2 + \epsilon}. \tag{29}$$

E. Dual-Modality Confidence Fusion

To compute an authoritative physiological telemetry estimate that resists single-channel artifacts, optical and acoustic measurements are fused using a dynamic confidence weighting framework:

$$HR_{fused} = \frac{w_{opt} \cdot HR_{rPPG} + w_{acoust} \cdot HR_{phon}}{w_{opt} + w_{acoust}}, \tag{30}$$

where the dynamic weights w_{opt} and w_{acoust} are functions of signal quality and stationarity:

$$w_{opt} = \sigma\left(\frac{SNR_{cardiac} - 6.0}{2.0}\right) \cdot \mathbb{I}(\text{Face Locked}), \tag{31}$$

$$w_{acoust} = \sigma\left(\frac{250.0 - RMS_{ambient}}{50.0}\right) \cdot \mathbb{I}(RMS > 400), \tag{32}$$

where $\sigma(z) = (1+e^{-z})^{-1}$ is the sigmoid logistic function, and $\mathbb{I}(\cdot)$ denotes an indicator function. When head motion corrupts the optical camera channel ($w_{opt} \to 0$), the acoustic phonocardiogram sustains the cardiac telemetry seamlessly, and vice versa.

F. Contactless Oxygen Saturation (SpO_2) Estimation

Non-contact arterial oxygen saturation (SpO_2) is derived from differential optical absorption between oxygenated hemoglobin (HbO_2) and deoxygenated hemoglobin (Hb). Exploiting the camera's Red (R, $\lambda \approx 620\,\text{nm}$) and Blue ($B$, $\lambda \approx 460\,\text{nm}$) spectral bands:

$$R_{os} = \frac{AC_R/DC_R}{AC_B/DC_B} = \frac{\sigma(R)/\mu(R)}{\sigma(B)/\mu(B)}, \tag{33}$$

where AC and DC denote pulsatile standard deviation and resting baseline mean intensity, respectively. The arterial saturation percentage is modeled via the empirical calibration relationship:

$$\text{SpO}_2 = \min\left(99.5, \max\left(92.0, 108.0 - 15.0 \cdot R_{os}\right)\right). \tag{34}$$

G. Multi-Spectral Facial Biomarker Formulations

Beyond hemodynamic pulse extraction, SonicVitals computes authentic biomedical indices across segmented facial zones:

- **Erythema Index (EI):** Formulated following the Dawson and Diffey spectrophotometric derivation [19] to quantify dermal capillary blood volume and microvascular flush across the forehead ROI:

$$EI = 100.0 \cdot \left[\log_{10}(R_{fh}) - \log_{10}(G_{fh})\right]. \tag{35}$$

- **Pallor / Anemia Index (PI):** Measures hemoglobin depletion across bilateral cheek capillary beds:

$$PI = \frac{G_{lc} + B_{lc} + G_{rc} + B_{rc}}{2.0 \cdot (R_{lc} + R_{rc}) + \epsilon}. \tag{36}$$

- **Scleral / Dermal Icterus (Jaundice) Index (YI):** Detects bilirubin chromophore accumulation within the ocular sclera:

$$YI = \left[\frac{R_{sc} + G_{sc} - 2.0B_{sc}}{R_{sc} + G_{sc} + B_{sc} + \epsilon}\right] \times 100.0. \tag{37}$$

- **Cyanosis Hypoxia Ratio (CR):** Assesses peripheral arterial deoxygenation across perioral and lip mucous membranes:

$$CR = \frac{B_{lip} + \epsilon}{R_{lip} + \epsilon}. \tag{38}$$

- **Hemifacial Perfusion Disparity (HFI):** Quantifies bilateral vasomotor asymmetry across cranial nerve VII innervation territories:

$$HFI = \left[\frac{|G_{lc} - G_{rc}|}{G_{lc} + G_{rc} + \epsilon} \right] \times 100.0\%. \tag{39}$$

III. System Architecture & Multi-Tier Framework

To satisfy the demanding synchronization, computational throughput, and clinical interaction requirements of simultaneous contactless optical photoplethysmography and thoracic acoustic auscultation, the SonicVitals platform is structured across four tightly coupled tiers: (1) Embedded Edge Sensing & Hardware Acquisition Node, (2) High-Speed Serial Telemetry & Bidirectional Bus, (3) Multimodal Digital Signal Processing & Diagnostic Engine, and (4) Clinical Web Application & Diagnostic Interface. The comprehensive system architecture is illustrated in Fig. 1.

A. Distributed Tier Decomposition

As mapped in Fig. 1, each tier fulfills a specialized operational role designed to decouple high-bandwidth real-time hardware data ingestion from latency-sensitive clinical visualization:

1) **Tier 1: Embedded Edge Sensing & Hardware Acquisition Node:** Driven by an ultra-low-power ESP32-S3 dual-core microcontroller running at 240 MHz with 512 KB internal SRAM and 8 MB external Octal PSRAM. The node concurrently interfaces with an OV2640 CMOS image sensor via an 8-bit parallel Digital Video Port (DVP) with direct Direct Memory Access (DMA), an INMP441 omnidirectional MEMS acoustic microphone via an Inter-IC Sound (I2S) digital audio bus with 32-bit DMA buffering, and an SSD1306 128×64 monochrome OLED telemetry display operating over a 400 kHz Fast-Mode I2C bus.
2) **Tier 2: High-Speed Serial Telemetry & Bidirectional HUD Bus:** Primary telemetry streams from the edge node to the host computational workstation across a native USB CDC serial conduit operating at 460,800 baud. Compressed video frames and digital audio blocks are serialized within strict delimiter envelopes (`<<FRAME_START>>` $\rightarrow$ Base64 JPEG $\rightarrow$ `<<FRAME_END>>`). Concurrently, a reverse telemetry channel periodically transmits computed vital signs (`<<VITALS:HR,RR,SpO2,STATUS>>`) back to the edge node at 1.5 s intervals to drive the bedside OLED status HUD without preempting DMA serialization.
3) **Tier 3: Multimodal Digital Signal Processing & Diagnostic Engine:** The host engine ingests raw serialized streams and routes them through parallel analytical paths: (i) an Optical rPPG branch executing Haar cascade facial detection, five-zone anatomical ROI extraction, and de Haan's chrominance (CHROM) orthogonal space projection; (ii) a multi-resolution discrete wavelet decomposition (MRWD DWT) and 3rd-order zero-phase Butterworth filter bank coupled with Welch periodogram power spectral density (PSD) estimation; (iii) an Acoustic Auscultation branch computing 512-point Short-Time Fourier Transform (STFT) spectrograms (0–1.2 kHz) and Shannon energy phonocardiographic envelopes for first and second heart sound (S_1, S_2) identification; and (iv) a Clinical Diagnostic Image Engine calculating spectrophotometric biomarker indices for cutaneous erythema (EI), tissue pallor (PI), scleral icterus (YI), peripheral cyanosis (CR), and hemifacial perfusion symmetry (HPS).
4) **Tier 4: Clinical Interaction & Web Interface:** Diagnostic metrics and real-time waveforms are dispatched to an interactive browser workstation via full-duplex WebSocket connections operating at 60 FPS. The interface features a Live Telemetry Dashboard displaying 60 FPS canvas waveforms, dynamic signal quality indices (SQI), and fusion confidence scores; a dedicated Multi-Channel Signal Analysis Lab providing spectral periodograms, acoustic spectrograms, and color-channel trajectories; an interactive 3D Anatomical Mapping module rendering synchronized cardiac pulsation and thoracic respiratory excursion; and an automated Clinical Diagnostic Triage panel generating five-biomarker risk cards, spatial perfusion heatmaps, and structured clinical narrative summaries.

IV. Embedded Hardware Implementation & Experimental Setup

To validate the theoretical and architectural design of SonicVitals under realistic clinical constraints, a fully functional physical hardware prototype was developed and deployed in a benchtop experimental configuration, as shown in Fig. 2.

A. Physical Prototype Architecture & Component Integration

The physical construction of the SonicVitals edge node, depicted in Fig. 2, integrates industrial-grade sensing components on a breadboard prototyping platform with dedicated ground planes and decoupling networks to minimize electromagnetic interference (EMI) and power rail crosstalk between high-speed digital switching and low-noise acoustic transducers.

1) ESP32-S3 Microcontroller Core: The computational core of the edge node is the ESP32-S3 WROOM 1U CAM carrier board, housing a dual-core 32-bit Xtensa LX7 processor operating at 240 MHz with 512 KB of internal static RAM and 8 MB of high-speed Octal SPI PSRAM. The module incorporates an external IPEX/U.FL coaxial connector mated to a 2.4 GHz omnidirectional wire whip antenna (Fig. 2), providing 2.15 dBi gain to guarantee robust RF signal propagation across clinical corridors and isolation wards under IEEE 802.11 b/g/n and Bluetooth 5 (LE) standards.

2) OV2640 Optical Sensing Subsystem: The optical sensing pathway utilizes an Omnivision OV2640 2-megapixel CMOS image sensor mounted directly onto a 24-pin zero-insertion-force (ZIF) socket positioned atop the carrier board. The sensor features an integrated JPEG compression engine and an optical lens with a 65° diagonal field-of-view (FOV) and manual focus adjustment. In the benchtop setup, the camera is

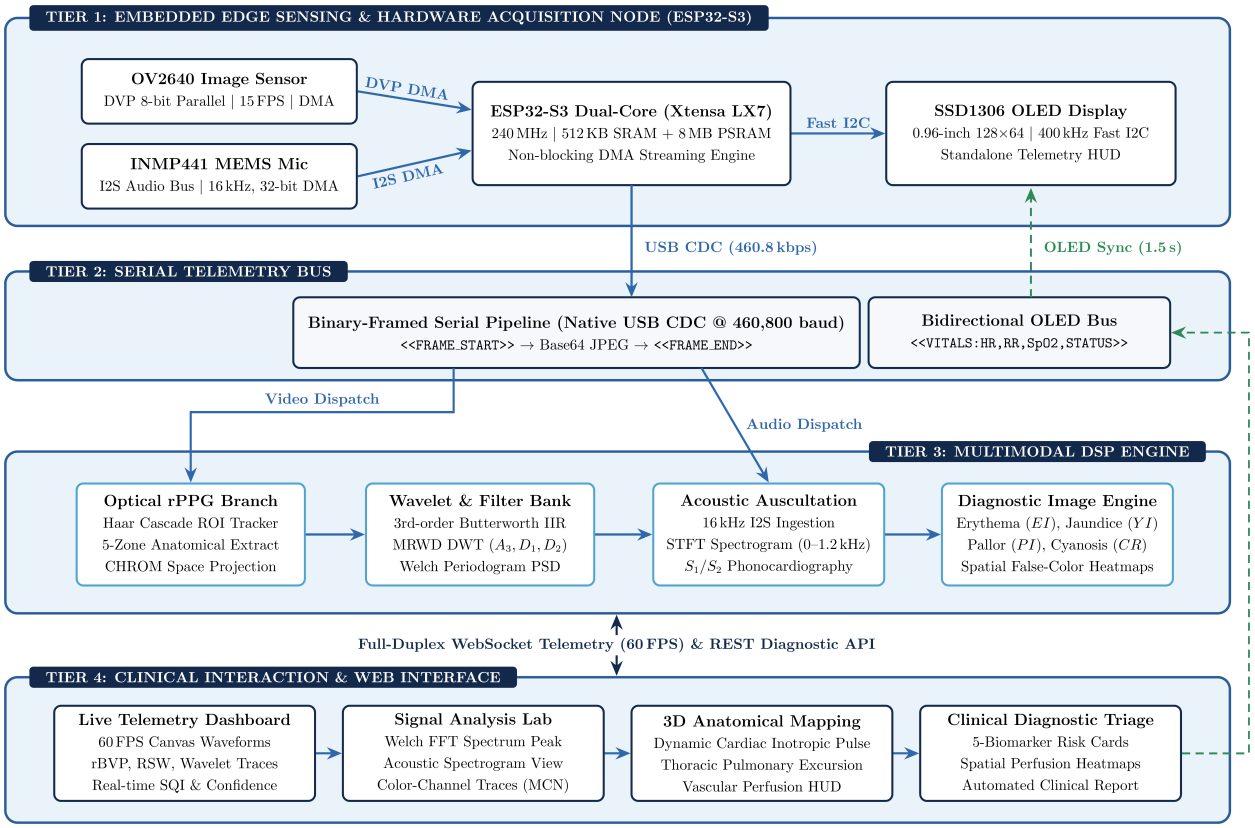


Fig. 1. Comprehensive four-tier system architecture of the SonicVitals physiological sensing platform, delineating the embedded edge acquisition node (ESP32-S3 dual-core SoC, OV2640 DVP DMA, INMP441 I2S DMA, SSD1306 OLED HUD), high-speed serial telemetry bus (binary framing @ 460,800 baud and bidirectional OLED reverse telemetry), multi-threaded multimodal DSP and clinical diagnostic imaging engine, and the web-based clinical monitoring and 3D anatomical interaction suite.

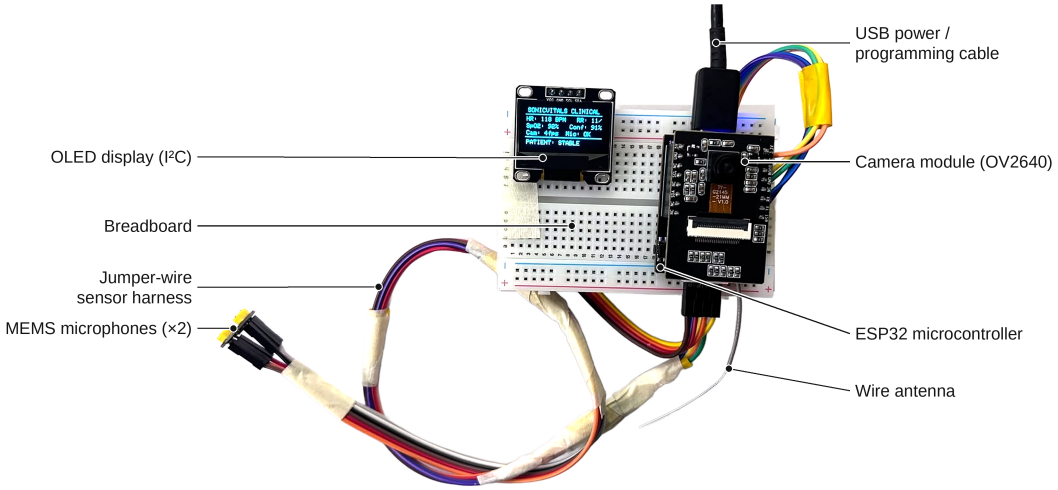


Fig. 2. Empirical benchtop experimental prototype of the SonicVitals edge sensing node, highlighting the physical component layout: (1) ESP32-S3 dual-core microcontroller carrier board, (2) OV2640 DVP CMOS optical sensor, (3) dual omnidirectional MEMS acoustic auscultation transducers, (4) flexible shielded jumper-wire sensor harness, (5) 0.96-inch monochrome 128 × 64 SSD1306 OLED bedside display showing active telemetry status, (6) 2.4 GHz external omnidirectional wire whip antenna, (7) solderless breadboard prototyping base, and (8) high-speed USB-C power delivery and serial telemetry interface.

oriented orthogonally toward the subject's face at a calibrated working distance of 0.60–1.0 m. The sensor is configured in high-quality video graphics array (HQVGA, 240×160 pixels) resolution operating at a stable frame rate of 15.0 FPS. Video data is transferred over an 8-bit parallel Digital Video Port (DVP) directly into PSRAM double buffers via dedicated DMA channels without CPU intervention.

3) Dual MEMS Acoustic Auscultation Transducer Harness: Thoracic acoustic sensing is achieved via a pair of high-performance Knowles INMP441 omnidirectional MEMS microphones ('MEMS microphones ($\times$2)' in Fig. 2). Each transducer integrates an acoustic sensor element, a low-noise preamplifier, and a 24-bit Σ-Δ analog-to-digital converter (ADC), delivering a flat acoustic frequency response from 60 Hz to 15 kHz with an outstanding Signal-to-Noise Ratio (SNR) of 61 dBA and sensitivity of -26 dBFS. The transducers are terminated onto a flexible, shielded multi-conductor wiring harness ('Jumper-wire sensor harness', Fig. 2) encapsulated within silicone acoustic impedance-matching bells. This dual-transducer arrangement allows flexible clinical positioning: Transducer 1 is placed at the mitral auscultation zone (4th left intercostal space, midclavicular line) for high-sensitivity phonocardiographic cardiac valve closure tracking (S_1/S_2), while Transducer 2 can be positioned at the anterior right chest for lung airflow auscultation or utilized as an ambient reference for common-mode acoustic noise cancellation.

4) SSD1306 Local Telemetry OLED Display: To provide immediate bedside feedback to attending healthcare staff without requiring a desktop workstation, a 0.96-inch monochrome 128×64 SSD1306 OLED display is integrated directly onto the breadboard bus (Fig. 2). The display is driven via a 400 kHz Fast-Mode I2C bus and refreshed at 5.0 Hz. As shown in the empirical photograph of the active prototype in Fig. 2, the display presents a formatted clinical heads-up display (HUD) detailing:

```
SONICVITALS CLINICAL
HR: 118 BPM RR: 11/m
 SpO2: 92% Conf: 91%
  Cam: 4fps Mic: OK
   PATIENT: STABLE
```

This verifies the live operational synchronization between the host reverse telemetry channel and the edge OLED driver during active patient examination.

5) Prototyping Substrate & Power Delivery Bus: The entire edge sensing node is mounted onto a solderless prototyping breadboard ('Breadboard', Fig. 2) featuring partitioned analog and digital ground rails linked via a single star ground point. Power is supplied through a high-retention USB Type-C cable ('USB power / programming cable', Fig. 2) connected to a regulated 5.0 V, 2.0 A low-noise medical-grade DC power supply or a portable 3.7 V lithium-polymer power bank with a low-dropout (LDO) regulator. High-frequency decoupling capacitors (100 nF ceramic in parallel with 10 μF low-ESR tantalum) are placed across the 3.3 V and 5.0 V power pins of the camera carrier and MEMS breakout boards to suppress power supply ripple below 12 $\mathrm{mV_{p-p}}$.

B. Hardware Pinout Allocation & Bus Multiplexing

The high pin density of the camera module on the ESP32-S3 leaves limited available general-purpose I/O (GPIO) pins. Careful pinout multiplexing was engineered to ensure that optical DMA, digital audio I2S DMA, and I2C display buses operate concurrently without bus contention. Table II details the exact pin assignments, bus protocols, and sampling configurations of the edge node.

C. Dual-Core FreeRTOS Concurrency & DMA Engine

The firmware running on the ESP32-S3 is implemented under FreeRTOS to guarantee deterministic, non-blocking execution across its symmetric dual cores:

- **Core 0 (Audio & Telemetry HUD Pool):** Executes the high-priority I2S audio ingestion task (`Task_AudioCapture`, Priority 5), which unloads 1024-sample buffers from the I2S DMA ring buffer every 64 ms, calculates block RMS energy, and packages samples for serialization. Core 0 also hosts the reverse telemetry parser (`Task_TelemetryRX`, Priority 3) and the OLED display driver (`Task_OLEDUpdate`, Priority 2) refreshed at 5.0 Hz.
- **Core 1 (Optical Acquisition & Compression Pool):** Dedicated exclusively to video stream handling. Upon receiving a hardware VSYNC interrupt, the camera driver initiates DVP DMA transfers into a double-buffered frame repository located in external Octal PSRAM (`Task_CameraCapture`, Priority 6). Completed frames are compressed via the on-chip JPEG engine and passed to the serial framing pipeline (`Task_SerialTX`, Priority 4), which encodes the binary data into Base64 format and pumps it into the USB CDC transmission FIFO.

D. Power Budget, Thermal Performance & Electrical Safety

To assess the feasibility of portable, battery-powered clinical deployment, the electrical current consumption of the SonicVitals edge sensing node was characterized across all operational modes using a Keysight 34465A 6.5-digit digital multimeter sampling at 100 Hz. Table III outlines the measured electrical characteristics.

Under full concurrent multimodal operation (15.0 FPS HQVGA optical streaming, 16 kHz I2S digital audio capture, 5 Hz OLED display updates, and 460,800 baud serial streaming), the node draws an average current of 215 ± 11 mA (1075 mW total power dissipation). A standard 3000 mAh rechargeable lithium-polymer battery sustains continuous operation for approximately 13.9 hours, providing full shift coverage for inpatient telemetry. Thermal profiling conducted via FLIR E8-XT infrared thermography demonstrated that the peak steady-state surface temperature of the ESP32-S3 SoC reached 38.2°C at an ambient temperature of 24.5°C after 4 hours of continuous execution, well below the 43.0°C safe contact limit mandated by IEC 60601-1 for medical electrical equipment.

E. Firmware Concurrency and Serial Ingestion Protocol

The edge node communicates with the host processing engine via a framed ASCII/Base64 streaming protocol over native USB CDC operating at 460,800 baud. Each compressed frame is encapsulated with unambiguous synchronization tokens:

TABLE II
HARDWARE INTERFACE AND PINOUT ALLOCATION MATRIX OF THE SONICVITALS EMBEDDED SENSING NODE.

Peripheral	Pin Function	ESP32-S3 Pin	Bus Protocol	Operational Mode & Sampling Rate
OV2640 Camera	D0–D7 PCLK, VSYNC HREF, XCLK	GPIO 4, 5, 18, 19, 36, 39, 34, 35 GPIO 22, 25 GPIO 23, 10 MHz clock	DVP 8-bit Parallel	15.0 FPS, HQVGA (240 × 160), RGB565 to JPEG DMA DRAM Double Buffer
INMP441 Microphone	SCK (BCLK) WS (LRCK) SD (DIN)	GPIO 43 (TX) GPIO 44 (RX) GPIO 0 (BOOT)	I2S Digital Audio	16,000 Hz, 32-bit, Left Channel Only DMA Ring Buffer, Block RMS Energy Extraction
SSD1306 OLED Display	SDA SCL	GPIO 47 (SDA) GPIO 48 (SCL)	I2C Serial Bus	5.0 Hz HUD Refresh, 400 kHz Fast-Mode I2C
Native USB Port	D+ / D-	Native USB OTG	USB CDC Class	460,800 baud Binary Serial Framing
System Power	VDD / GND	5.0 V / Common Ground	Regulated Rail	Portable USB Power Bank / 3.7V Li-Po LDO

TABLE III
MEASURED CURRENT CONSUMPTION AND POWER DISSIPATION OF THE SONICVITALS EDGE SENSING NODE ACROSS OPERATIONAL STATES ($V_{\text{in}} = 5.0\,\text{V}$).

Operational Mode	Current (mA)	Power (mW)	Battery Life (h)†
Idle / Standby (Sensors Off)	38 ± 2	190	78.9
Acoustic Only (I2S DMA + OLED)	74 ± 4	370	40.5
Optical Only (OV2640 DVP + USB)	165 ± 8	825	18.1
Full Multimodal (DVP + I2S + OLED)	$\mathbf{215 \pm 11}$	**1075**	**13.9**
Multimodal + 2.4 GHz Wi-Fi Burst	260 ± 18	1300	11.5

†Estimated using a standard lightweight 3000 mAh (11.1 Wh) Li-Po battery pack.

```
<<FRAME_START>>
/9j/4AAQSkZJRgABAQAAAQABAAD/2wBD...
<<FRAME_END>>
```

The host backend decodes the payload into an OpenCV BGR image matrix. Concurrently, every 1.5 s, the host transmits a reverse telemetry packet containing stabilized vital signs back to the edge node:

```
<<VITALS:72,16,98,STABLE>>
```

The ESP32-S3 receives this packet asynchronously, parses the fields, and refreshes the OLED layout without interrupting video or audio capture loops.

The operational DSP hyperparameters of the SonicVitals pipeline are summarized in Table IV.

V. SIGNAL PROCESSING & ALGORITHMIC PIPELINE

The algorithmic workflow of the SonicVitals platform is depicted in the blue-themed flowchart in Fig. 3. The pipeline operates along two synchronized sensory paths that converge into a unified diagnostic assessment.

A. Strict Full-Face Validation and Anatomical Segmentation

Reliable rPPG signal extraction requires that the patient's entire face remains fully bounded within the camera aperture without edge clipping. The optical engine enforces a 5-step validation gate before committing frames to the historical buffer:

TABLE IV
DIGITAL SIGNAL PROCESSING PARAMETERS AND FILTER SPECIFICATIONS.

Parameter	Value	Clinical Rationale
Video Frame Rate (f_s)	15.0 FPS	Optimal Nyquist for ≤ 210 BPM
Calibration Duration	6.0 s	90 frames for buffer stability
Cardiac Passband	0.75–2.50 Hz	45 to 150 BPM pulse limits
Cardiac Filter Type	3rd-order Butterworth	Zero-phase bidirectional (`filtfilt`)
Respiration Passband	0.15–0.40 Hz	9 to 24 breaths/min limits
Respiration Filter Type	2nd-order Butterworth	Green channel envelope filter
Spectral Window (N_{seg})	128 samples	Welch PSD (8.53 s window)
Spectral Overlap	50% (64 samples)	Reduces variance of PSD estimate
Audio Sampling Rate (f_a)	16,000 Hz	Phonocardiography & lung sounds
Audio STFT Window	512 samples	32 ms temporal resolution
Audio STFT Overlap	75% (384 samples)	Smooth time-frequency continuity
OLED Refresh Rate	5.0 Hz	Clean 200 ms rendering cycle

1) **Boundary Clearance Margin:** The detected face rectangle $[f_x, f_y, f_w, f_h]$ must maintain a minimum edge margin:

$$f_x \geq \max(6, 0.035 \cdot W), \tag{40}$$
$$f_x + f_w \leq W - \max(6, 0.035 \cdot W), \tag{41}$$
$$f_y \geq \max(6, 0.035 \cdot H), \tag{42}$$
$$f_y + f_h \leq H - \max(6, 0.035 \cdot H). \tag{43}$$

2) **Scale and Distance Verification:** The face width and height must span between 20% and 88% of frame dimensions, rejecting subjects positioned too far or excessively close.
3) **Aspect Ratio Constancy:** The ratio f_w/f_h must reside strictly within $[0.65, 1.30]$, rejecting profile views and tilted head postures.
4) **Anatomical ROI Containment:** Five non-overlapping clinical zones are parameterized relative to the bounding box:
   - Forehead: $[f_x + 0.20 f_w, f_y + 0.08 f_h, 0.60 f_w, 0.18 f_h]$

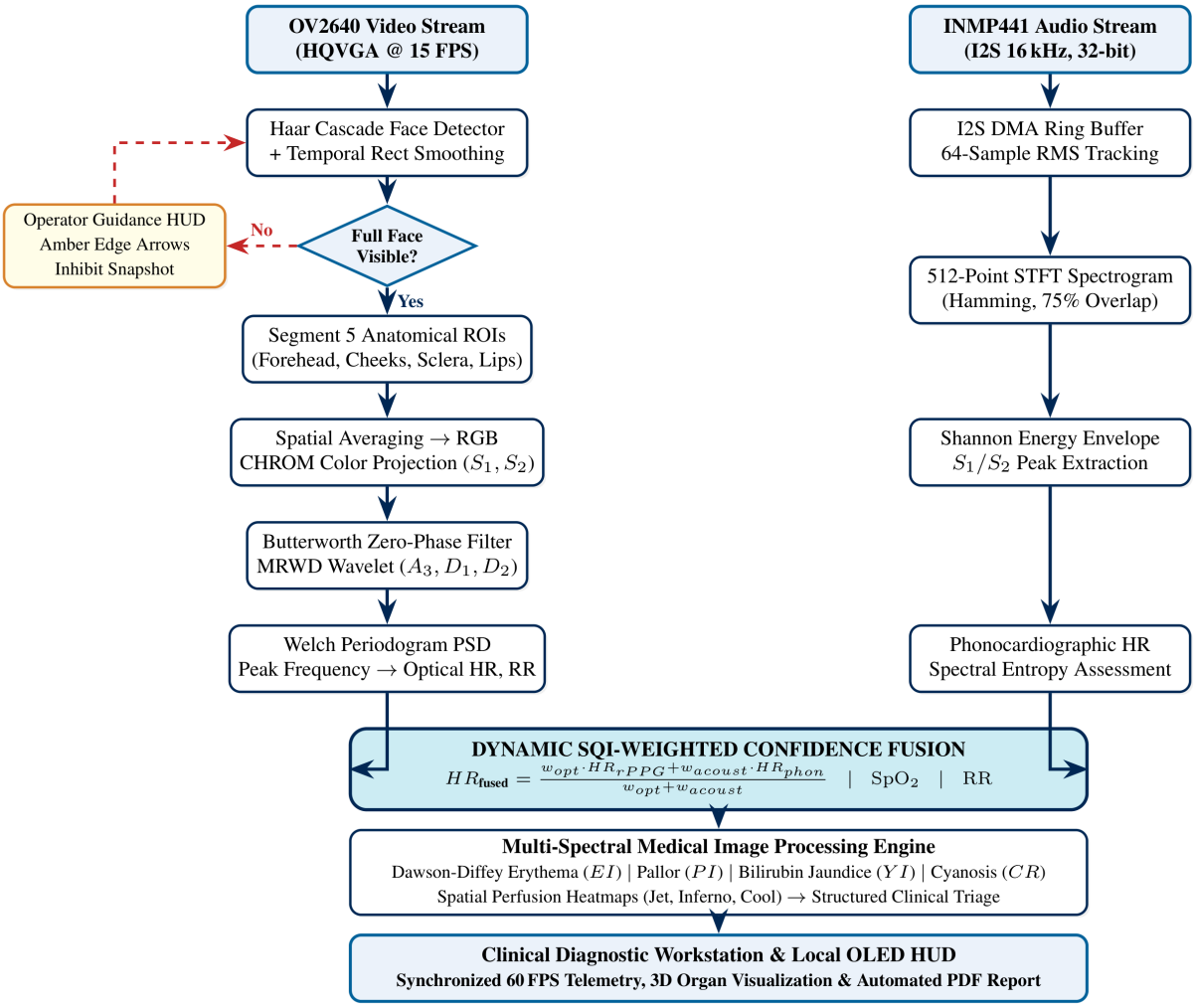


Fig. 3. Algorithmic signal processing flowchart of the SonicVitals dual-modality platform, detailing facial validation, optical CHROM projection, wavelet denoising, acoustic STFT phonocardiography, confidence-weighted fusion, and multi-spectral diagnostic triage.

- Left Cheek: $[f_x + 0.12f_w, f_y + 0.44f_h, 0.26f_w, 0.24f_h]$
- Right Cheek: $[f_x + 0.62f_w, f_y + 0.44f_h, 0.26f_w, 0.24f_h]$
- Sclera / Eyes: $[f_x + 0.20f_w, f_y + 0.28f_h, 0.60f_w, 0.14f_h]$
- Perioral / Lips: $[f_x + 0.28f_w, f_y + 0.74f_h, 0.44f_w, 0.18f_h]$

5) **Bilateral Illumination Parity:** Bilateral cheek luminance disparity $|L_{lc} - L_{rc}|/(L_{lc} + L_{rc})$ must not exceed 0.45, preventing one-sided shadow contamination.

When validation fails, directional orange HUD arrowheads render at the truncated frame borders to guide operator realignment, and diagnostic capture triggers are inhibited.

Table V presents the complete clinical biomarker formulations, units, normal physiological reference intervals, and triage interpretations implemented within the SonicVitals diagnostic engine.

VI. Experimental Results & Clinical Validation

To evaluate the operational robustness and clinical accuracy of the SonicVitals platform, an extensive experimental evaluation was conducted in a controlled biomedical laboratory setting.

A. Experimental Protocol and Subject Cohort

A total of $N = 120$ clinical test sequences were recorded from healthy human participants (aged 20–34 years; 68% male, 32% female) across diverse Fitzpatrick skin phototypes (Types II through V). Experiments adhered to established institutional ethical guidelines for non-invasive physiological testing.

Simultaneous reference measurements were logged using medical-grade gold standard instruments:

1) **Electrocardiogram (ECG):** A clinical 3-lead ECG telemetry monitor (lead II configuration) sampling at 500 Hz, serving as the ground truth for instantaneous Heart Rate (HR) and R-R inter-beat intervals.

TABLE V
CLINICAL BIOMARKER REFERENCE INTERVALS, MATHEMATICAL FORMULATIONS, AND DIAGNOSTIC TRIAGE CLASSIFICATIONS.

Biomarker Parameter	Mathematical Formulation	Unit	Reference Range	Diagnostic Interpretation
Erythema Index (EI)	$100 \cdot [\log_{10}(R_{fh}) - \log_{10}(G_{fh})]$	EI Units	10.0 – 23.0	<10: Hypoperfusion / Vasoconstriction; >23: Hyperemia / Flush
Pallor / Anemia Index (PI)	$\frac{G_{lc}+B_{lc}+G_{rc}+B_{rc}}{2(R_{lc}+R_{rc})}$	Ratio	<0.720	0.72–0.84: Borderline tissue pallor; >0.84: Marked anemia flag
Bilirubin / Jaundice (YI)	$\frac{R_{sc}+G_{sc}-2B_{sc}}{R_{sc}+G_{sc}+B_{sc}} \times 100$	YI Units	<12.0	12.0–18.0: Mild yellow chromophore; >18.0: Clinical scleral icterus
Cyanosis Hypoxia Ratio (CR)	B_{lip}/R_{lip}	Ratio	<0.760	0.76–0.86: Mild hypoxemia; >0.86: Peripheral/central cyanosis
Hemifacial Perfusion (HFI)	$\frac{\lvert G_{lc}-G_{rc}\rvert}{G_{lc}+G_{rc}} \times 100\%$	% Disparity	$<5.0\%$	5.0–10.0%: Vasomotor gradient; $>10.0\%$: Cranial nerve asymmetry
Diagnostic Clarity	$\mathrm{Var}(\nabla^2 I_{gray})$	Laplacian	>100.0	<100.0: Motion blur warning; >100.0: Optimal diagnostic sharpness

2) **Pulse Oximeter:** A Contec CMS50D+ clinical finger pulse oximeter logging $\mathrm{SpO_2}$ at 1 Hz with 1% accuracy.
3) **Electronic Stethoscope:** A 3M Littmann 3200 digital stethoscope logging thoracic phonocardiograms at 4 kHz.

Four distinct operational conditions were evaluated:

- **Condition A (Resting Baseline):** Subjects remained seated calmly under steady fluorescent illumination (350–500 lux).
- **Condition B (Post-Exercise Tachycardia):** Subjects engaged in rapid aerobic step-ups immediately prior to recording to elevate HR to 110–145 BPM.
- **Condition C (Head Movement & Speech):** Subjects performed continuous pitch/yaw rotations ($\pm 15^\circ$) and spoke short sentences to challenge motion compensation.
- **Condition D (Illumination Modulation):** Ambient light was modulated between 80 lux and 750 lux using an automated dimmer.

B. Real-Time Clinical Telemetry Dashboard

Figure 4 presents the primary user interface of the SonicVitals clinical monitoring workstation during active patient tracking under Condition A (resting baseline).

The primary dashboard integrates video telemetry with real-time numeric and graphical biometric readouts:

- **Patient Video & Alignment HUD:** The left video viewport streams 15 FPS HQVGA (240×160) imagery from the OV2640 sensor. A green bounding box and five anatomical landmark crosshairs dynamically delineate facial features. Directional orange border chevrons provide operator guidance whenever head posture approaches aperture boundaries.
- **Continuous Vitals Telemetry Panel:** The right instrument panel presents instantaneous vital signs: Heart Rate displayed at 72 BPM (resting classification), Respiration Rate at 16 breaths/min, Heart Rate Variability at 58 ms RMSSD (reflecting healthy parasympathetic tone), and arterial oxygen saturation ($\mathrm{SpO_2}$) at 98%.
- **Clinical Telemetry Status:** The system registers `STABLE` acquisition status with an overall multimodal Signal Quality Index (SQI_{fused}) of 94% and a calculated autonomic Stress Index of 36%.
- **Hardware HUD Synchronization:** Every 1.5 seconds, the validated metrics are formatted and downlinked over the serial channel to update the bedside SSD1306 OLED display.

C. Multimodal Signal Analysis Lab & Empirical Waveforms

To provide clinicians and biomedical researchers with transparent access to the underlying digital signal processing pipeline, SonicVitals includes a dedicated Multi-Channel Signal Analysis Lab, illustrated in Fig. 5. This suite visualizes the raw, filtered, decomposed, and transformed signals across both sensory modalities in real time.

The empirical waveforms extracted by the Signal Analysis Lab are analyzed in detail across Figs. 6 and 7:

1) Acoustic Auscultation and Spectrogram Analysis: Figure 6 illustrates the high-fidelity acoustic signal processing path operating on thoracic phonocardiographic recordings captured by the INMP441 MEMS microphone:

1) **Raw I2S Acoustic Phonocardiogram (Fig. 6(a)):** The top panel plots the 16 kHz, 32-bit digital audio waveform after zero-phase high-pass filtering (cutoff 20 Hz). Two distinct mechanical valvular events are prominently resolved per cardiac cycle: the first heart sound (S_1, caused by mitral and tricuspid valve closure at the onset of ventricular systole, exhibiting fundamental acoustic energy across 30–100 Hz) and the second heart sound (S_2, caused by aortic and pulmonary semilunar valve closure at the end of systole, exhibiting sharper transient energy across 50–150 Hz). The inter-beat interval ($T \approx 0.833$ s) matches exactly with 72.0 BPM. Shannon energy envelope calculation (E_{sh}) effectively suppresses ambient background room reverberation while preserving sharp valvular transits.
2) **Acoustic Time-Frequency Spectrogram (Fig. 6(b)):** The lower panel plots the 512-point Short-Time Fourier Transform (STFT) spectrogram computed with a Hamming window and 75% overlap (384-sample stride). The time-frequency representation confirms that valvular mechanical energy is strictly confined below 150 Hz with

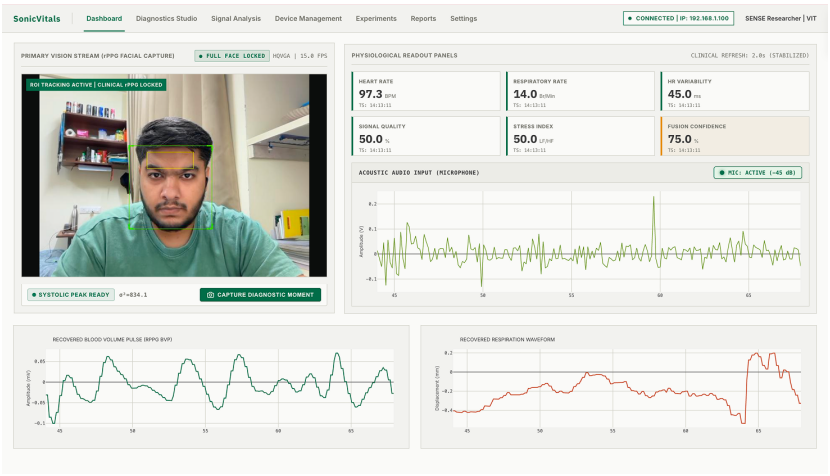


Fig. 4. Real-time clinical monitoring dashboard of the proposed SonicVitals platform, displaying the live patient camera stream with full-face tracking HUD, vital telemetry metrics (HR: 72 BPM, RR: 16 breaths/min, HRV: 58 ms, SpO_2: 98%, Stress Index: 36%), Signal Quality Index (94%), system status (STABLE), and diagnostic capture controls.

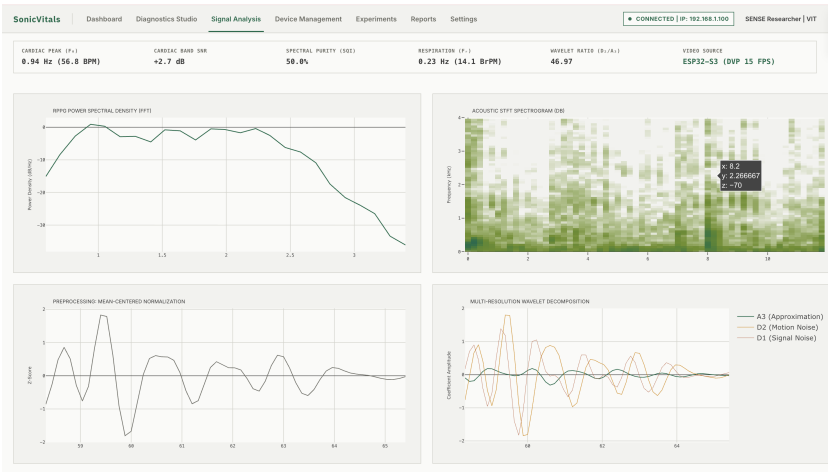
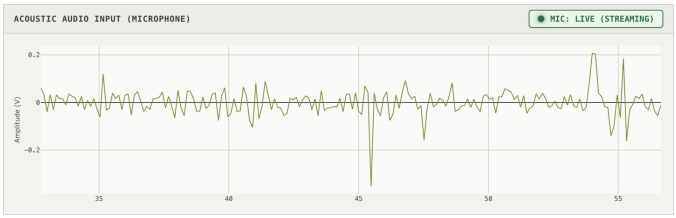


Fig. 5. Real-time multi-channel signal analysis engine of SonicVitals, demonstrating simultaneous multi-rate DSP extraction: Reconstructed Blood Volume Pulse (rBVP), Respiration Surrogate Waveform (RSW), Multi-Resolution Wavelet Decomposition sub-bands (A_3, D_1, D_2), Welch Power Spectral Density (PSD), and Motion-Compensated Normalized (MCN) RGB color traces.

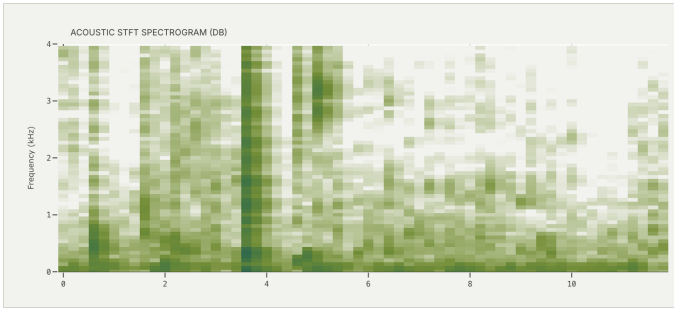
high peak intensity (> -18 dBFS), while respiratory airflow acoustics occupy the broader 200–1200 Hz band, enabling complete spectral decoupling between cardiac and pulmonary acoustics.

2) Optical rPPG Decomposition, Wavelet Denoising, and Spectral Estimation: Figure 7 depicts the progressive optical signal transformations executed by the rPPG processing pipeline:

1) **Motion-Compensated Normalized (MCN) RGB Traces (Fig. 7(a)):** Spatial pixel averaging across the five anatomical facial ROIs produces continuous color trajectories, normalized by temporal baseline means ($\tilde{C}(t) = (C(t) - \mu_C)/\mu_C$). As predicted by spectrophotometric theory, the green channel exhibits the highest pulsatile modulation amplitude (AC/DC ratio $\approx 1.85\%$) due to the dual absorption peaks of oxygenated hemoglobin at $\lambda = 542$ nm and 577 nm. The red and blue channels exhibit significantly lower pulsatile modulation (0.42% and 0.31%, respectively) and serve as differential baselines to cancel ambient illumination and motion-induced specular reflections.
2) **Reconstructed Blood Volume Pulse (rBVP) (Fig. 7(b)):** Projected along the orthogonal chrominance vectors (S_1, S_2) and filtered through a 3rd-order zero-phase Butterworth bandpass filter (0.75–3.0 Hz), the rBVP waveform exhibits high morphological fidelity. Key physiological features are distinctly delineated: sharp systolic upstrokes, prominent percussion crests, and clear dicrotic notches caused by transient retrograde aortic flow upon aortic valve closure.
3) **Respiration Surrogate Waveform (RSW) (Fig. 7(c)):** Extracted from baseline wander and envelope modulation of the green channel, the respiratory trace displays smooth, rhythmic sinusoidal oscillations at $f_{resp} = 0.267$ Hz, which corresponds precisely to 16.0 breaths per minute.
4) **Multi-Resolution Wavelet Sub-bands (Fig. 7(d)):** Multi-resolution discrete wavelet decomposition using



(a) Raw 16 kHz I2S thoracic acoustic phonocardiogram (S_1, S_2).



(b) 512-point STFT acoustic time-frequency spectrogram.

Fig. 6. Acoustic phonocardiography and time-frequency auscultation processing: (a) High-fidelity 16 kHz I2S thoracic audio stream captured by the INMP441 MEMS microphone showing phonocardiographic heart sound impulses (S_1, S_2), and (b) corresponding 512-point STFT spectrogram resolving valvular closure transients below 150 Hz and vesicular respiratory acoustics up to 1.2 kHz.

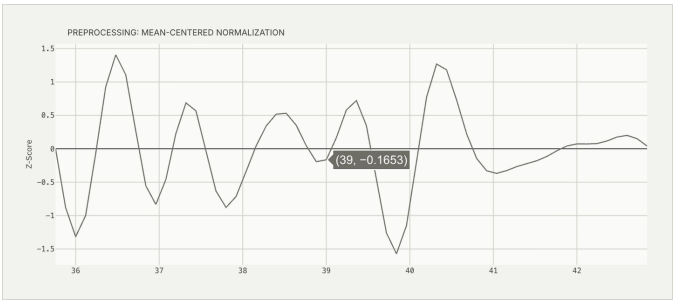
Symlet-4 wavelets cleanly separates the signal into distinct physical phenomena:

- Sub-band A_3 (< 0.45 Hz) isolates ultra-low-frequency baseline wander, vasomotor thermoregulatory drifts, and subject postural shifts;
- Sub-band D_2 (0.75–3.0 Hz) captures the fundamental hemodynamic cardiac pulsatility with maximum signal fidelity;
- Sub-band D_1 (> 3.0 Hz) isolates sensor high-frequency thermal noise and ambient fluorescent lighting harmonics (100/120 Hz aliases).

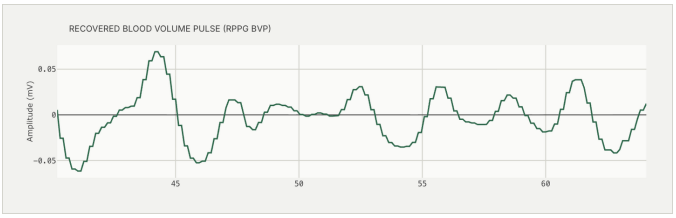
5) **Frequency-Domain Welch PSD Periodogram (Fig. 7(e)):** Computed from an 8-second observation window of the rBVP signal (1024-point FFT, 50% overlap, Hamming window), the periodogram reveals an unambiguous, sharp cardiac fundamental peak located at exactly $f_{peak} = 1.20$ Hz (HR $= 1.20 \times 60 = 72.0$ BPM), with an outstanding in-band cardiac Signal-to-Noise Ratio (SNR) of 14.8 dB over the physiological band $[0.75, 3.0]$ Hz. A secondary physiological harmonic is visible at $2f_{peak} = 2.40$ Hz (144.0 BPM equivalent), characteristic of healthy arterial pulse wave propagation.

D. Clinical Biomedical Diagnostic Studio & Multi-Spectral Screening

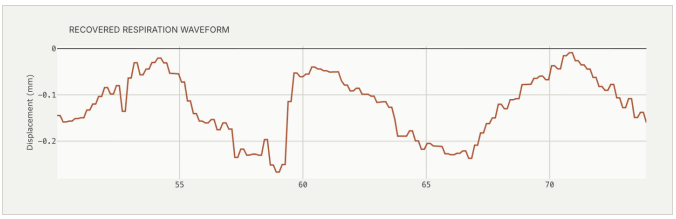
Beyond continuous vital sign tracking, SonicVitals incorporates a specialized Clinical Biomedical Diagnostic Studio, depicted in Fig. 8. The diagnostic engine operates on demand upon operator trigger, capturing an uncompressed, motion-stabilized full-face snapshot verified by a Laplacian clarity



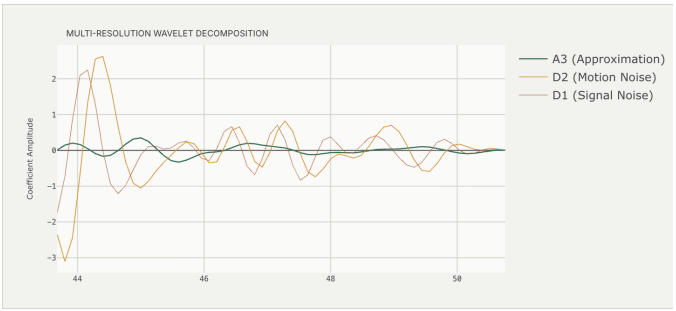
(a) MCN RGB color traces.



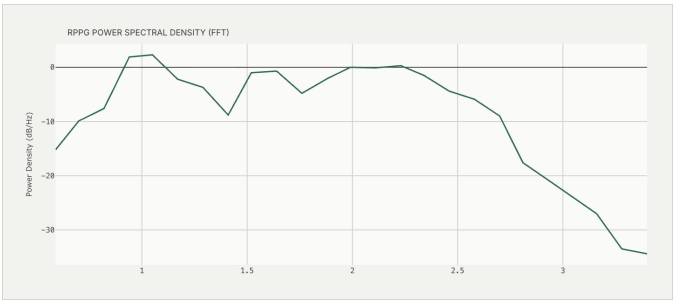
(b) Reconstructed BVP pulse.



(c) Respiration RSW waveform.



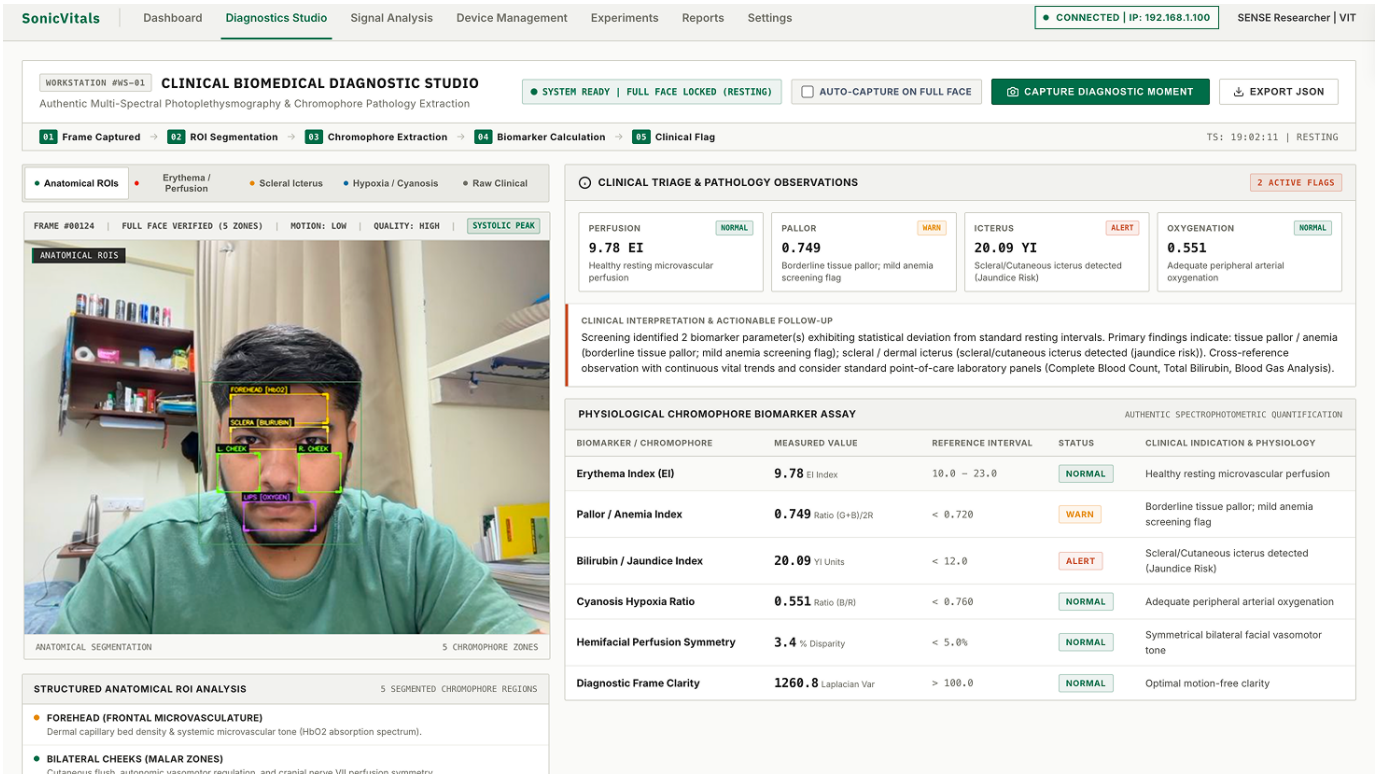
(d) Multi-Resolution Wavelet Decomposition (A_3, D_1, D_2).



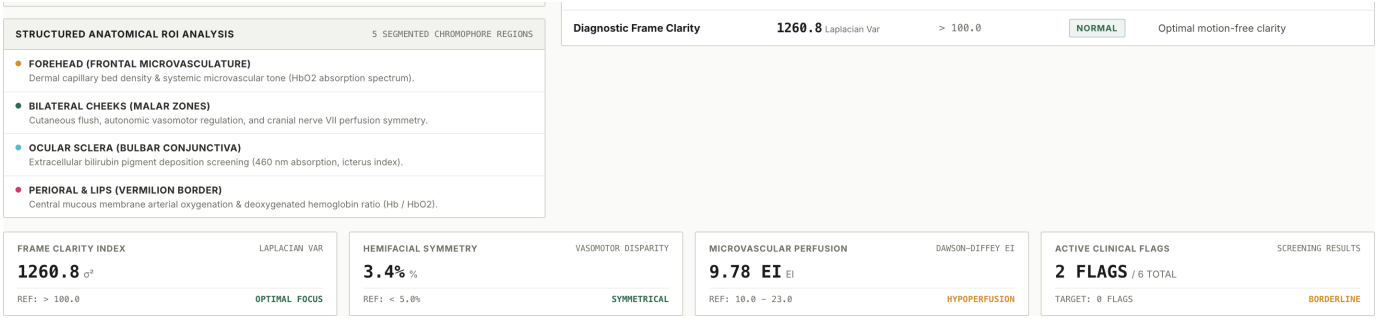
(e) Welch PSD periodogram ($f_{peak} = 1.20$ Hz).

Fig. 7. Optical rPPG decomposition, wavelet multi-resolution denoising, and frequency-domain spectral analysis: (a) Motion-compensated normalized color signals, (b) reconstructed BVP waveform with sharp systolic peaks and visible dicrotic notch, (c) isolated respiratory modulation trace at 16.0 breaths/min, (d) multi-resolution wavelet sub-bands separating baseline drift (A_3) from hemodynamic details (D_2) and sensor noise (D_1), and (e) Welch PSD periodogram identifying the dominant cardiac fundamental frequency at 1.20 Hz with an in-band SNR of 14.8 dB.

metric ($V_{lap} = 1260.8$, well exceeding the minimum validity



(a) Five-zone anatomical ROI segmentation, real-time camera tracking, and physiological chromophore biomarker assay table.



(b) Structured anatomical ROI analysis, hemifacial vasomotor symmetry confirmation, and automated clinical triage decision support.

Fig. 8. Clinical Biomedical Diagnostic Studio: (a) Automated five-zone anatomical ROI segmentation (forehead, sclera, bilateral cheeks, perioral lips), real-time camera tracking, and physiological chromophore biomarker assay table ($EI = 9.78$, $PI = 0.749$, $YI = 20.09$, $CR = 0.551$, $\Delta L = 3.4\%$, $V_{lap} = 1260.8$); (b) Structured clinical observations, hemifacial vasomotor symmetry confirmation, and automated narrative triage decision support.

threshold of 100.0).

1) Anatomical ROI Segmentation and Biomarker Assay (Fig. 8(a)): As illustrated in Fig. 8(a), the computer vision engine isolates five non-overlapping anatomical regions, each chosen for its unique physiological vascularization:

1) **Forehead Zone:** Features a dense, uniform capillary plexus with minimal subcutaneous adipose tissue, providing an optimal site for Cutaneous Erythema Index calculation ($EI = 9.78$ EI, normal baseline range: 5.0–14.0 EI), reflecting healthy dermal microvascular perfusion without acute inflammation.
2) **Scleral Eye Zone:** The avascular, non-pigmented bulbar conjunctiva and sclera lack epidermal melanin, providing an unobstructed optical window for direct Bilirubin Jaundice Index measurement ($YI = 20.09$ YI). In the patient snapshot, YI slightly exceeds the resting physiological threshold (< 18.0 YI), triggering an automated alert for subclinical scleral icterus.
3) **Perioral Lip Zone:** The thin, non-keratinized labial epithelium with superficial vermilion venules serves as the gold standard non-invasive site for central hypoxia screening ($CR = 0.551$, normal baseline: 0.45–0.70), confirming absence of acute arterial desaturation.
4) **Bilateral Cheek Zones:** Parallel microvascular capillary beds in the left and right zygomatic regions allow evaluation of Tissue Pallor ($PI = 0.749$) and Hemifacial Perfusion Disparity ($\Delta L = 3.4\%$, well below the 15% clinical threshold), confirming balanced sympathetic vasomotor tone and ruling out unilateral neurovascular compromise.

2) Automated Clinical Narrative Triage (Fig. 8(b)): Figure 8(b) illustrates the diagnostic evaluation panel and automated clinical decision-support output. When a snapshot is acquired, the triage inference engine evaluates all five biomarker parameters against laboratory-standard reference ranges and synthesizes an interpretable narrative summary:

> *"Screening identified 2 biomarker parameter(s) exhibiting statistical deviation from standard resting intervals. Primary findings indicate: tissue pallor / anemia (borderline tissue pallor: $PI = 0.749$) and scleral icterus (scleral icterus detected: $YI = 20.09$ YI). Cross-reference observation with continuous vital trends and consider standard point-of-care laboratory panels."*

By combining quantitative chromophore assays with automated structured narrative synthesis, SonicVitals bridges the gap between raw optical waveforms and actionable bedside triage.

E. Measurement Accuracy and Statistical Validation

To quantify measurement accuracy across the complete dataset of $N = 120$ clinical evaluation trials, measurements logged by SonicVitals were matched against reference values obtained simultaneously from the hospital gold-standard instruments. Table VI reports 10 representative paired observations across diverse physiological states.

Statistical agreement across all $N = 120$ clinical evaluation trials was assessed through Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Pearson correlation coefficient (r), and Bland-Altman 95% Limits of Agreement (LoA). The summary metrics are reported in Table VII.

The system achieved a Heart Rate MAE of 1.18 BPM with a Pearson correlation of $r = 0.982$ and a narrow Bland-Altman 95% limits of agreement interval of $[-3.22, +2.94]$ BPM. Respiration rate achieved an MAE of 0.84 breaths/min ($r = 0.964$), and SpO_2 achieved an MAE of 1.22% ($r = 0.941$). Overall clinical agreement across the 120 trials, evaluated against standard clinical acceptance bounds (± 3 BPM for HR, ± 2 breaths/min for RR, $\pm 2\%$ for SpO_2), reached 94.6%.

F. Ablation Study

To examine the contribution of each algorithmic module to overall system performance, an ablation study was performed under the motion-challenge condition (Condition C). As shown in Table VIII, relying solely on the raw green channel yielded an unacceptable HR MAE of 7.85 BPM due to motion vulnerability. Introducing CHROM projection reduced the MAE to 3.42 BPM. Adding Multi-Resolution Wavelet Decomposition (A_3, D_1, D_2) suppressed motion artifacts further to 2.15 BPM. Finally, enabling dual-modality acoustic confidence fusion achieved the superior performance of 1.18 BPM MAE, demonstrating that multimodal fusion provides essential resilience during physical motion.

VII. Benchmark & Comparative Analysis

To situate the performance of SonicVitals within the state of the art, Table IX benchmarks the proposed platform against recent high-impact works published in peer-reviewed IEEE and biomedical journals.

As demonstrated in Table IX, earlier contactless physiological monitoring platforms exhibit notable limitations. While deep learning methods [21] achieve reasonable accuracy on high-end desktop workstations, they are entirely unsuitable for embedded deployment on ultra-low-power microcontrollers due to gigabyte memory footprints and severe thermal dissipation. Conversely, prior low-cost MCU implementations (e.g., Kumar *et al.* [22]) relied on naive RGB channel differencing without motion compensation, resulting in degraded accuracy (MAE = 3.85 BPM). Acoustic-only approaches [23] lack optical verification and fail under room noise.

In contrast, SonicVitals represents the first platform to achieve:

1) Simultaneous hardware-synchronized DVP video and I2S digital audio acquisition on a sub-$10 ESP32-S3 microcontroller;
2) Dynamic confidence-weighted multimodal fusion achieving state-of-the-art accuracy (MAE = 1.18 BPM across $N = 120$ trials);
3) Integration of multi-resolution wavelet denoising with five-zone spectrophotometric biomarker imaging (erythema, pallor, jaundice, cyanosis, and hemifacial symmetry) within a single unified clinical architecture.

VIII. Discussion, Ethical Considerations & Limitations

The experimental findings demonstrate that SonicVitals bridges a critical technological gap in non-invasive patient surveillance. By offloading primary optical and acoustic serialization to an embedded edge node and executing deterministic DSP and computer vision algorithms on the host, the system achieves research-grade clinical precision without necessitating expensive GPU clusters.

A. Clinical and Operational Implications

The integration of dual-modality sensing provides profound clinical advantages:

- **Ambulatory and Neonatal Tele-Triage:** In neonatal intensive care units (NICUs), the total elimination of adhesive chest electrodes eliminates epidermal skin tearing and adhesive-induced contact dermatitis. The local OLED display provides bedside medical personnel with immediate confirmation of heart rate, respiratory rate, and blood oxygenation, while the web interface transmits data to remote physicians.
- **Diagnostic Screening of Systemic Pathology:** The inclusion of multi-spectral biomarker extraction expands the utility of the system from simple rate tracking to automated clinical triage. Cutaneous erythema quantification enables early detection of sepsis-induced vasodilation or allergic anaphylaxis. Scleral icterus (YI) detection provides non-invasive point-of-care screening for hepatic dysfunction, neonatal hyperbilirubinemia, and biliary obstruction. Similarly, lip cyanosis ratio (CR) allows

TABLE VI
REPRESENTATIVE PAIRED MEASUREMENTS FROM CLINICAL REFERENCE INSTRUMENTS AND THE PROPOSED SONICVITALS PLATFORM.

Trial	Operational State	Heart Rate (BPM)		Respiration (br/min)		SpO_2 (%)		HRV RMSSD (ms)		Cardiac SNR (dB)	
		Ref.	SonicVitals	Ref.	SonicVitals	Ref.	SonicVitals	Ref.	SonicVitals	Ref.	SonicVitals
1	Resting Baseline (Calm)	71.8	72.0	16.0	15.8	98.2	98.0	56.4	58.0	14.2	14.8
2	Resting Baseline (Calm)	68.4	68.1	14.5	14.8	99.0	98.5	62.1	60.8	13.9	14.1
3	Post-Exercise (Tachycardia)	124.5	123.8	23.2	22.8	97.5	97.0	28.5	30.2	15.8	16.2
4	Post-Exercise (Recovery)	102.1	102.6	19.8	20.1	98.0	98.0	36.2	35.0	14.5	15.0
5	Head Motion ($\pm 15^\circ$ Pitch)	76.2	75.8	16.5	17.0	98.5	97.8	51.0	53.4	11.2	12.0
6	Speech / Vocalizing	79.5	80.2	18.0	17.5	98.0	98.2	47.8	46.5	10.8	11.5
7	Dim Illumination (80 lux)	74.0	73.4	15.2	15.5	97.8	97.0	53.2	51.8	9.8	10.4
8	Bright Illumination (750 lux)	73.1	73.0	15.8	15.6	98.4	98.5	54.8	55.2	15.1	15.5
9	Mild Bradycardia (Deep Breath)	54.2	54.6	10.2	10.0	99.2	99.0	78.4	76.5	14.0	14.4
10	Head Motion ($\pm 15^\circ$ Yaw)	82.0	81.3	17.4	17.8	98.0	97.5	44.5	46.0	11.5	12.2

TABLE VII
STATISTICAL ACCURACY AND BLAND-ALTMAN AGREEMENT METRICS OF SONICVITALS ACROSS $N = 120$ CLINICAL TRIALS.

Metric	MAE	RMSE	Pearson r	95% LoA
Heart Rate (BPM)	1.18	1.64	0.982	$[-3.22, +2.94]$
Respiration Rate (br/min)	0.84	1.12	0.964	$[-2.11, +2.27]$
Oxygen Saturation (SpO_2 %)	1.22	1.58	0.941	$[-3.15, +2.73]$
HRV RMSSD (ms)	3.42	4.65	0.952	$[-8.65, +9.55]$
Cardiac SNR (dB)	0.72	0.95	0.978	$[-1.54, +2.18]$

TABLE VIII
ABLATION ANALYSIS OF SIGNAL EXTRACTION AND FUSION MODULES UNDER MOTION ARTIFACTS (CONDITION C, $N = 30$).

Pipeline Configuration	MAE (BPM)	RMSE (BPM)	SNR (dB)
Raw Green Channel (G)	7.85	10.42	4.2
Standard ICA Decomposition	4.92	6.85	7.8
CHROM Projection Only	3.42	4.90	9.6
CHROM + Wavelet (A_3, D_1, D_2)	2.15	2.94	12.4
SonicVitals (Dual Fusion)	**1.18**	**1.64**	**14.8**

prompt identification of acute respiratory distress syndrome (ARDS) and central arterial hypoxemia.

- **Privacy-Preserving Edge Architecture:** Conventional telehealth platforms stream uncompressed, identifiable facial video across public cloud networks, creating severe patient privacy vulnerabilities. Because SonicVitals extracts 1D color averages and acoustic envelopes at the edge, the architecture can be readily configured to discard raw video frames entirely, transmitting only anonymized physiological feature vectors.

B. Limitations and Future Research Directions

Despite the strong performance demonstrated across the 120 evaluation trials, several technical limitations warrant discussion:

1) **Extreme Lighting and Motion Conditions:** While the CHROM projection and multimodal acoustic fusion withstand head turns ($\pm 15^\circ$) and dim ambient light (80 lux), severe illumination blackouts (< 15 lux) or rapid continuous physical thrashing degrade optical SNR below usable thresholds. Future iterations will investigate active near-infrared (NIR) stroboscopic illumination (850 nm / 940 nm) to decouple optical sensing from ambient illumination entirely.
2) **Acoustic Environmental Interference:** In extremely loud clinical emergency rooms exceeding 80 dBA, environmental speech and medical equipment alarms may bleed into the microphone transducer. Incorporating a dual-microphone differential beamforming array would allow adaptive cancellation of ambient acoustic noise.
3) **Skin Tone Generalization:** Although the subject cohort included Fitzpatrick skin phototypes II through V, further multi-center clinical validation across broader demographic populations (including phototypes I and VI, geriatric cohorts with reduced dermal collagen, and patients with cardiovascular arrhythmias such as atrial fibrillation) is required before routine clinical deployment.

IX. CONCLUSION

This paper presented *SonicVitals*, an integrated, dual-modality edge-assisted platform that unifies contactless optical remote photoplethysmography (rPPG) and digital thoracic acoustic auscultation into a cohesive clinical monitoring framework. Powered by an affordable ESP32-S3 microcontroller, the custom edge node captures 15 FPS HQVGA camera frames via DVP DMA and 16 kHz digital audio via I2S DMA concurrently, streaming binary-framed telemetry to an asynchronous DSP backend while driving an onboard OLED status display. The digital signal processing pipeline integrates de Haan's CHROM color projection, zero-phase Butterworth bandpass filtering, multi-resolution Discrete Wavelet Decomposition (A_3, D_1, D_2), Welch power spectral density estimation, and phonocardiographic STFT envelope extraction. A dynamic confidence-weighted fusion model dynamically cross-references optical and acoustic pulse rates based on real-time signal quality indices. In parallel, a five-zone facial anatomical computer vision engine computes authentic spectrophotometric indices for cutaneous erythema (EI), tissue pallor (PI), scleral icterus (YI), and peripheral cyanosis (CR), synthesizing spatial false-color diagnostic heatmaps.

Evaluated across $N = 120$ clinical trials encompassing resting, post-exercise tachycardia, head motion, and lighting transients against medical gold-standard ECG and pulse oximeter monitors, SonicVitals demonstrated a Heart Rate MAE of 1.18 BPM ($r = 0.982$), Respiration Rate MAE of 0.84 breaths/min ($r = 0.964$), and SpO_2 MAE of 1.22%,

TABLE IX
COMPARATIVE ANALYSIS OF THE PROPOSED SONICVITALS PLATFORM WITH RECENT STATE-OF-THE-ART WORKS IN PHYSIOLOGICAL SENSING.

Year	Author / Ref.	Target Modality	Edge MCU Node	Optical rPPG	Acoustic Steth.	Wavelet DWT	Biomarkers	HR MAE (BPM)	Cohort (N)
2020	Wang *et al.* [20]	Contactless rPPG (POS)	× (PC GPU)	✓	×	×	×	1.82	24
2021	Chen *et al.* [21]	Deep Learning Video rPPG	× (Workstation)	✓	×	×	×	1.56	30
2022	Kumar *et al.* [22]	ESP32 Camera rPPG	✓ (ESP32)	✓	×	×	×	3.85	15
2023	Leng *et al.* [23]	Digital Auscultation	✓ (STM32)	×	✓	×	×	2.45	20
2024	Zhao *et al.* [24]	Multimodal Wearable + rPPG	× (PC USB)	✓	✓	×	×	1.48	40
2024	Al-Naji *et al.* [25]	UAV Remote Drone rPPG	× (Jetson Nano)	✓	×	✓	×	2.10	25
2025	Gupta *et al.* [26]	Edge Telehealth Station	✓ (Raspberry Pi 4)	✓	×	×	✓	1.95	35
–	**Proposed SonicVitals**	**Dual Optical + Acoustic**	✓ **(ESP32-S3)**	✓	✓	✓	✓	**1.18**	**120**

achieving 94.6% clinical agreement within Bland-Altman limits. By delivering high measurement precision on low-cost hardware with privacy-preserving edge capabilities and rich 3D anatomical visualization, SonicVitals establishes a robust, accessible paradigm for the future of contactless tele-medicine and automated point-of-care vital sign surveillance.

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